

SmolVLM

Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University | [arxiv:2504.05299](#)

256M / 500M / 2.2B

Model Sizes

< 1 GB RAM

Smallest Model

300×

Larger than Llafix-80B
(which SmolVLM-256M surpasses)

Large VLMs Are Powerful but Impractical for Edge Deployment

The memory wall blocking on-device multimodal AI

The Problem

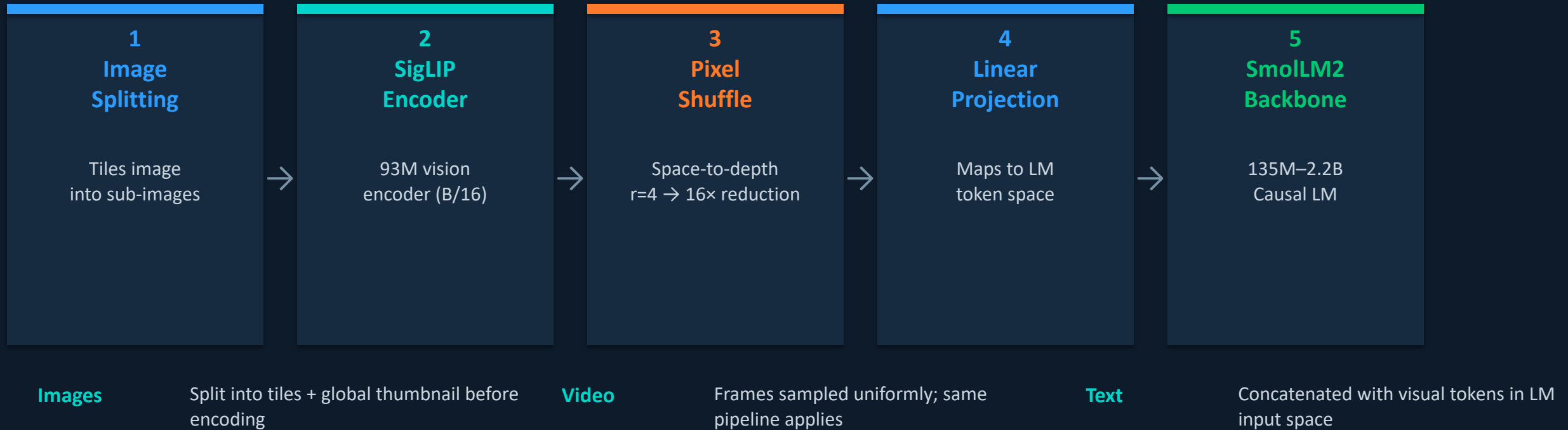
- State-of-the-art VLMs (e.g., GPT-4V, LLaVA-34B) require 20–80 GB GPU RAM
- Cloud inference adds latency, cost, and privacy risks
- Mobile devices offer only 4–16 GB shared RAM
- Prior small VLMs sacrificed too much accuracy to be useful
- No model previously ran sub-1 GB inference with competitive scores

Memory Requirements

Model	RAM
Idefics-80B	160+ GB
Molmo-A1B-7B	27.7 GB
InternVL2-2B	~5 GB
SmolVLM-2.2B	4.9 GB
SmolVLM-500M	1.2 GB
SmolVLM-256M	0.8 GB

SmolVLM Architecture: Image Splitting, Pixel Shuffle & SmolLM2

End-to-end pipeline from raw image to language model tokens



Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1 — Balanced encoder-LM parameter allocation matters

Key Finding:

A 93M SigLIP-B/16 encoder paired with a 135M LM outperforms a 428M SigLIP-SO400M with the same LM.

Only at 1.7B LM scale does the larger encoder justify its 10% additional parameter overhead.

Rule of thumb:

Match encoder capacity to LM capacity.
Over-investing in vision while keeping a tiny LM wastes parameters.

Encoder Size vs. Performance (135M LM Scale)

Relative Score (%)

100%

82%

101%

93M
SigLIP-B/16

428M
SigLIP-SO400M

428M
SigLIP-SO400M
@ 1.7B LM

Better

Worse at
small LM scale

Justified at
large LM scale

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16x for Small Models

Finding 3 — Space-to-depth transformation balances compression vs. accuracy

How Pixel Shuffle Works

Spatial features are rearranged into channel dimension:

$$H \times W \times C \rightarrow (H/r) \times (W/r) \times (C \cdot r^2)$$

At r=4:

Tokens reduced by factor $r^2 = 16\times$

Drastically lowers attention sequence length

Enables processing of more image tiles per budget

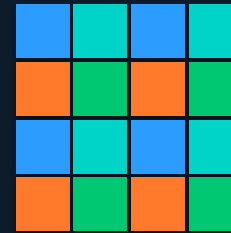
Tradeoff:

Larger models $\rightarrow r=2$ (4x compression) preserves detail

Small models $\rightarrow r=4$ (16x compression) is optimal

The attention savings outweigh any quality loss for compact LMs

Token Count Reduction by Shuffle Factor



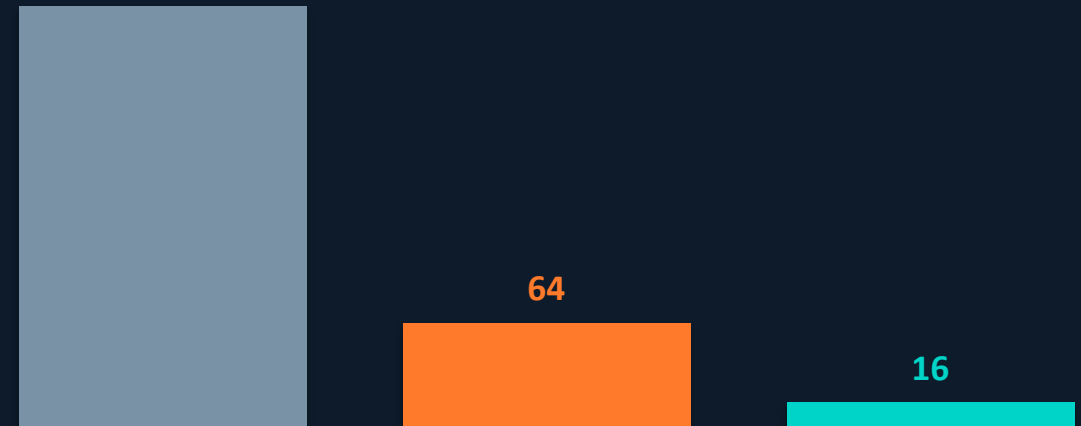
Input: 4x4 spatial grid

$\rightarrow r=2 \rightarrow$



Output: 2x2, 4x channels
(r=2: 4x fewer tokens)

Visual tokens per tile (normalized to r=1 = 256 tokens)



Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2 — RoPE base adjustment + mixed-context fine-tuning

Why Context Matters

- More context = more image tiles processable simultaneously
- At 2K tokens: ~4 tiles at r=4 (limited resolution/detail)
- At 16K tokens: ~60+ tiles → richer spatial understanding
- Critical for DocVQA, ChartQA, and dense-text benchmarks

How SmolVLM Extends Context

- RoPE positional encoding base: 10K → 273K
- Enables length generalization beyond training context
- Fine-tuned on mixed long/short-context data
- SmolVLM-2.2B → 16K tokens
- SmolVLM-256M/500M → 8K tokens

Impact on Benchmarks

- DocVQA: benefits most from long-context (80.0% at 2.2B)
- ChartQA: structured layout needs full-image context
- Video-MME: longer context → more frames per inference
- Performance ↑ monotonically from 2K to 16K in experiments

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Evaluation across 13 benchmarks — Single-Image, Multi-task, Video

Category	Benchmark	256M	500M	2.2B	Best Efficient OS	RAM
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo-A1B-7B)	—
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo-A1B-7B)	—
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo-A1B-7B)	—
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo-A1B-7B)	—
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo-A1B-7B)	—
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo-A1B-7B)	—
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo-A1B-7B)	—
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo-A1B-7B)	—
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo-A1B-7B)	—
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2-2B)	—
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2-2B)	—
AVERAGE	All Tasks	44.0%	51.0%	59.8%	—	0.8/1.2/4.9 GB

Molmo-A1B-7B requires 27.7 GB RAM | Green = SmolVLM beats baseline | SmolVLM-256M surpasses 300x larger Idefix-80B

Real-Time Inference on Apple Silicon: Up to 80 Tokens/Second

HuggingSnap iOS app — fully on-device, no cloud dependency

~80 tok/s

256M on Pro M1 Max 64GB
(batch=1)

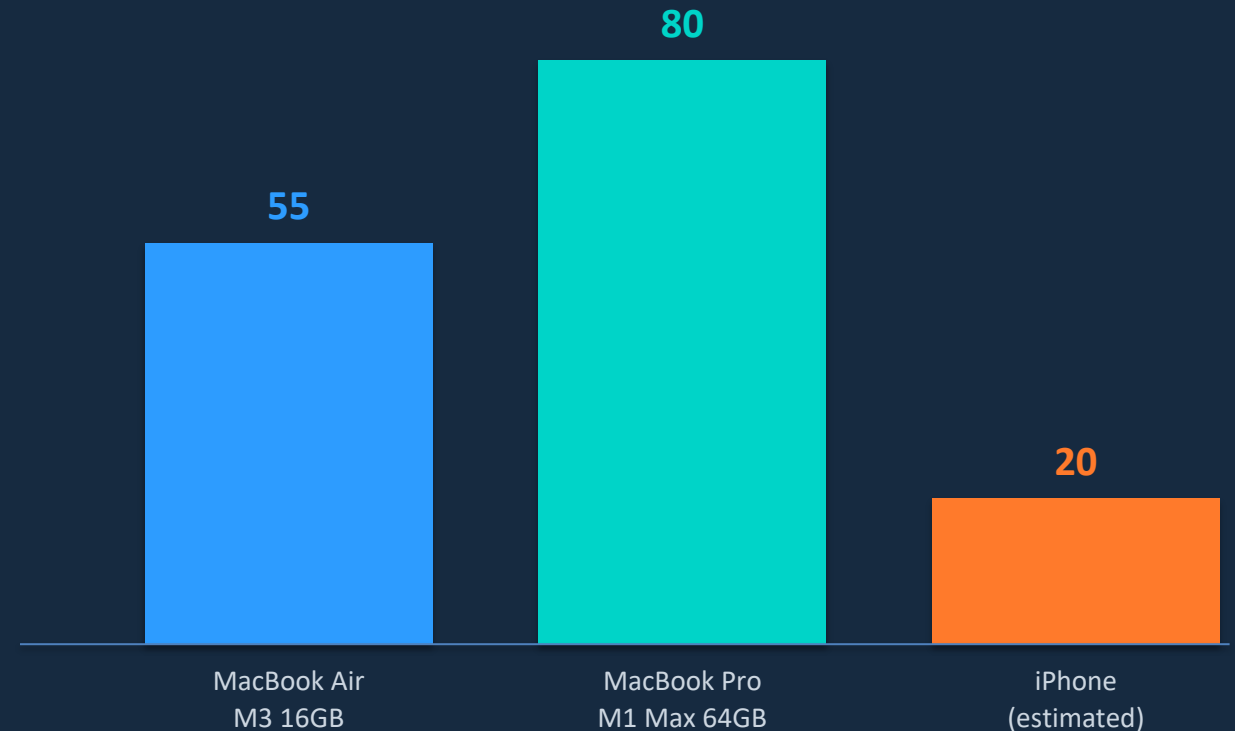
~55 tok/s

256M on MacBook Air M3 16GB
(batch=1)

~1000 tok/s

256M at batch=64 on high-end
Apple Silicon

Tokens / Second (SmolVLM-256M, batch=1)



HuggingSnap: open-source iOS app processing photos & videos entirely on-device. No cloud. No latency. Full privacy.

SmolVLM — Hugging Face & Stanford | arxiv:2504.05299

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Open Source Everything

Release weights, data, code, and HuggingSnap accelerates community progress. Reproducibility is a first-class design goal.

Actionable guidance for on-device AI research

1

Balance Encoder-LM Compute

Match vision encoder size to LM capacity. Over-parameterizing the encoder wastes compute — a 93M encoder beats 428M at small LM scales.

SmolVLM proves that thoughtful architecture beats raw scale — and open-sourcing democratizes on-device multimodal AI.

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